

A MAIN PROJECT REPORT ON

NEXTGENISP ERP & CRM

IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE AWARD OF THE DEGREE IN

BACHELOR OF COMPUTER APPLICATIONS

OF

MAHATMA GANDHI UNIVERSITY

KOTTAYAM

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DEPARTMENT OF COMPUTER APPLICATIONS

(2023-2026)

M E S COLLEGE , MARAMPALLY

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ALUVA -7

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DEPARTMENT OF COMPUTER APPLICATIONS

Certificate

This to certify that the report entitled

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In partial fulfillment of the award of the degree in

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1.

2.

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1. INTRODUCTION

1.1 Overview of the System

NextGenISP is a ERP and CRM platform designed specifically for the unique operational needs of Internet Service Provider (ISP) businesses. Unlike generic management tools, NextGenISP serves as a centralized "digital nervous system" that unifies all business functions from the moment a lead is captured to the ongoing technical maintenance of an active connection. By leveraging a high-performance Django REST Framework backend and a responsive React (Vite) frontend, the system provides a real-time, single source of truth for all business resources, including customer subscriptions, staff logistics, financial records, and physical equipment. The platform's ERP capabilities are realized through its tightly integrated functional modules. Meanwhile, the Automated Billing and Finance module eliminates revenue leakage by generating precise, recurring invoices and offering secure online payment options via integrated payment gateways. To ensure technical reliability, the Service & CRM module implements a smart ticketing system that distinguishes between logical and physical issues, ensuring that the right personnel are dispatched to the right problem at all times.

1.2 Problem Definition and Objective

The ISP industry frequently faces operational bottlenecks due to fragmented data and siloed internal systems. Without a unified ERP solution, billing cycles often fall out of sync with actual service status, potential leads are lost in manual spreadsheets, and support teams lack clear visibility into the physical location or serial numbers of customer equipment. The primary objective of NextGenISP is to resolve these critical operational pain points by providing an all-encompassing, data-driven ERP ecosystem. The system aims to automate the entire customer lifecycle, from initial enquiry through to field installation and ongoing subscription management, ensuring that financial and technical operations remain in lockstep.

2. REQUIREMENT AND ANALYSIS

2.1 Problem Definition

Many local Internet Service Providers currently operate using highly fragmented and manual administrative processes that hinder their ability to scale and provide reliable service. These operations often suffer from significant data siloing, where customer billing, technical support, and physical inventory are managed through disconnected tools such as manual spreadsheets or paper-based logs. This lack of integration leads to frequent data redundancy, human errors, and critical delays in service delivery. Furthermore, without a unified link between subscription status and billing cycles, businesses often experience significant revenue leakage and high administrative overhead for manual payment verification. Support teams are also hindered by a lack of real-time visibility into customer equipment, such as MAC addresses or serial numbers, which complicates troubleshooting and extends the time required to resolve technical issues. The absence of a centralized asset tracking system further exacerbates these problems, leading to high rates of hardware loss and an inability to proactively manage network outages through geospatial visualization.

2.2 Select the Software Development Model

The Agile Development Model was selected as the most appropriate methodology for developing the NextGenISP system. This model was chosen primarily for its iterative approach, which is essential for building a platform composed of diverse and complex modules such as GIS mapping, automated billing engines, and inventory management ensuring that each module meets high quality standards before being integrated into the main ERP hub. This flexibility is particularly valuable given the project's reliance on evolving third-party integrations, such as the payment gateway and geospatial mapping APIs, which may require technical adjustments as the implementation progresses.

2.3 Requirement Specification includes Existing and Proposed system

The existing systems in place for many service providers are largely manual or only semi-automated, relying on phone calls for lead capture and manual invoice generation. These systems often result in disconnected billing cycles where payments are collected via cash or bank transfers without automated verification, and technical support is managed through paper-based ticketing with no automated staff assignment. In contrast, the NextGenISP proposes a fully integrated, automated workflow that manages the customer lifecycle from the initial public enquiry through to lead verification, field installation, and active subscription. The proposed system justifies its implementation by centralizing all critical business data into a single Django/React ecosystem, ensuring complete data integrity across billing, support, and inventory functions. This unified approach provides substantial benefits, including a significant reduction in manual data entry overhead, automatic subscription suspension for non-payment to protect revenue, and the ability to use GIS-based visualization for localized maintenance alerts. By ensuring that billing and service delivery remain in constant synchronization, the proposed system allows for seamless scalability as the ISP grows.

2.3.1 Justification of the Proposed System

The implementation of the NextGenISP is justified by the critical need to bridge the operational gap between customer expectations and technical service delivery. In today's competitive environment, an ISP cannot thrive if its billing cycles are decoupled from its service activations, or if its support teams lack the real-time data required to resolve issues efficiently. By centralizing all business logic—including automated financial processing and geospatial asset tracking—into a secure digital ecosystem, the project ensures complete data integrity across all business functions.

2.3.2 Benefits of the Proposed system

The transition to the NextGenISP provides a wide array of tangible benefits that improve both the organization's financial health and the overall customer experience. From an operational standpoint, the automation of the onboarding and billing processes leads to a dramatic reduction in manual administrative labor, freeing up staff to focus on higher-value service activities. Financially, the system protects revenue by automatically suspending services for non-payment and providing a secure, transparent payment environment through integrated online gateways. For technical teams, the introduction of smart ticketing and GIS-based network mapping enables a shift from reactive to proactive maintenance, resulting in localized outage alerts and significantly faster response times.

2.4 Project Planning

The project has 5 months from November to March. Considering the total available time for the project, the following plan and schedule was prepared:

SI NO	Duration	Activity
1	November	Identification of need.
2	December	Feasibility study
3	December	Analysis
4	January	Design
5	February	Testing
6	March	Implementation

2.5 Project Scheduling

Activity	November	December	January	February
Identification of need				
Feasibility study				
Analysis				
Design				
Testing				
Implementation				

2.6 Feasibility Study

The feasibility of the project was analyzed across technical, economic, operational, and legal dimensions to ensure its long-term viability. Technical feasibility is confirmed by the use of mature frameworks like Django and React, which provide the robustness and flexibility needed for complex ERP functions. Economic feasibility is demonstrated by the use of open-source technologies that minimize initial licensing costs, while the long-term gains from automated billing and reduced staff overhead provide a strong return on investment. Operational feasibility is ensured by the platform's role-specific dashboard design, which aligns with existing ISP workflows and minimizes the need for extensive user retraining. Legal and Social feasibility is addressed through the implementation of strict data privacy measures and secure authentication protocols, ensuring that the system complies with modern data protection.

3. SOFTWARE REQUIREMENT SPECIFICATION(SRS)

3.1 Introduction

The Software Requirement Specification (SRS) document defines the functional and non-functional requirements of the NextGenISP system. It serves as a detailed blueprint for the system's design, development, and implementation. The SRS outlines the system's purpose, scope, features, user roles, and technical specifications to ensure smooth functionality and efficient project execution. By clearly defining the requirements, the SRS helps developers, designers, and stakeholders understand the system's objectives, ensuring that the final product meets user needs while maintaining performance, security, and scalability.

3.1.1 Purpose

The purpose of this Software Requirement Specification (SRS) is to provide a detailed and unambiguous description of the software requirements for the NextGenISP system. This document serves as a comprehensive guide for developers, project managers, and stakeholders to ensure that the system's functionalities, performance standards, and user interfaces align with the operational needs of an Internet Service Provider (ISP) business. By detailing the functional and non-functional requirements, this specification aims to minimize technical revisions during the development phase and provide a clear baseline for final system verification and deployment against predefined success criteria.

3.1.2 Scope

The scope of NextGenISP encompasses a complete (ERP) and (CRM) solution specifically tailored for the internet service provider industry. The system is designed to manage the entire customer lifecycle, ranging from initial lead generation and field installation to active subscription maintenance and technical support. real-time inventory and hardware asset tracking. The platform is not intended to provide the actual physical internet bandwidth, but rather to serve as the administrative.

3.1.3 Definitions, Acronyms, and Abbreviations

The NextGenISP project utilizes several industry-standard technical terms and acronyms to describe its various business and technical components. An Enterprise Resource Planning (ERP) system refers to the integrated management of core business processes, while an Internet Service Provider (ISP) denotes the entity delivering internet access to customers. The technical architecture is built on the Django REST Framework (DRF) for the specialized backend API and a Single Page Application (SPA) model for a responsive and modern frontend experience. Key technical terms include the Optical Network Terminal (ONT), which refers to hardware installed at the customer's site, and the JSON Web Token (JWT) used for secure, stateless authentication. Furthermore, GIS (Geographic Information System) is used to describe the geospatial mapping used in service zone visualization, and the Fair Usage Policy (FUP) defines the monthly data limits within individual internet plans.

3.1.4 References

- These primary references include the official technical documentation for the Django and React frameworks, as well as the standard IEEE 830-1998 guidelines for software requirement specifications.

3.1.5 Overview

This SRS document is structured to provide a logical and comprehensive flow from the objectives to the specific technical requirements of the NextGenISP. Following the document provides a detailed overall description of the system, the critical integration points between its core financial, logistical, and technical modules. Sections provide a granular breakdown of the specific functional requirements for lead management, billing, support. The final sections of the document are dedicated to non-functional requirements such as system security, ensuring that the system is not only functional but also scalable and secure for enterprise-level use.

3.2 Overall Description

NextGenISP is a specialized, independent ERP & CRM solution built to unify the disparate operational silos of a modern internet service provider. At its core, the platform serves as a central administrative hub that bridges the gap between customer requests, technical troubleshooting, and field logistics. The system's primary functionality revolves around automating the entire customer lifecycle—from the initial marketing lead to subscription activation and ongoing maintenance. This is achieved through role-specific dashboards tailored for administrators, technical staff, field personnel, and customers. The platform is designed to operate within modern web browser environments while adhering to strict data privacy standards through secure, role-based access controls. Its success depends on the reliable uptime of integrated gateway APIs and the consistent data entry of hardware serials by field staff, ultimately aiming to turn a complex, manual ISP operation into a streamlined and proactive digital ecosystem.

3.2.1 Product Perspective

NextGenISP is developed as an independent, comprehensive ERP solution designed to replace fragmented legacy systems within the ISP industry. While it functions as a standalone platform, it is engineered to integrate seamlessly with external services such as the payment gateway for financial transactions and various geospatial mapping APIs for network visualization. The system acts as the central management hub, interacting with a distributed network of hardware components, including routers and ONTs at customer locations, to provide a unified administrative interface. It is built using a modern decoupled architecture, allowing the React-based frontend to remain responsive and independent of the core Django-based business logic, thus ensuring that the system can be easily scaled or integrated with newer technologies as they emerge.

3.2.2 Product Functions

The primary functions of the NextGenISP system revolve around automating the diverse operational workflows required by a modern service provider. This includes a robust user onboarding engine that captures public enquiries and manages their progression through lead verification and field installation. The system also automates the core financial cycle by generating recurring invoices, processing online payments, and managing customer subscriptions in real-time. For technical operations, the platform provides a specialized CRM for ticket management, differentiating between physical and logical network issues and facilitating intelligent staff assignment. Furthermore, the system offers advanced inventory management to track hardware assets and integrated GIS mapping to provide visual insights into network coverage and customer distribution across various geographic zones.

3.2.3 User Characteristics

The system caters to four primary user groups, each with distinct technical backgrounds and operational needs. Administrators are high-level users who require a comprehensive overview of the business, including financial reports, staff performance metrics, and network health monitoring. Technical Staff members are specialized users responsible for network configuration and logical troubleshooting, requiring deep access to ticket details and customer hardware data. Field Staff are mobile users focused on physical infrastructure, such as line installations and hardware repairs, and they interact with the system primarily through a task-oriented, mobile-responsive interface. Finally, Customers are the end-users of the service who use the system to manage their personal profiles, view bills, make payments, and communicate with support through a simplified, intuitive portal.

3.2.4 Constraints

The development and operation of the NextGenISP system are subject to several technical and environmental constraints. From a technical standpoint, the system's performance is partially dependent on the reliability of the underlying SQLite or PostgreSQL database and the stability of third-party APIs for payments and mapping. The application must also operate within the security constraints of modern web browsers and maintain compatibility with standard React and Django development environments. Additionally, the system must navigate data privacy regulations, ensuring that sensitive customer information is stored and accessed through secure, role-based protocols. Hardware-related constraints also apply, as the system's ability to track assets relies on the accurate entry of MAC addresses and serial numbers by field personnel during the installation process.

3.2.5 Assumptions and Dependencies

The successful implementation of the NextGenISP depends on several key assumptions and external dependencies. It is assumed that the ISP has a reliable internal network infrastructure and that customers have access to a stable internet connection to interact with the web-based portal. The system is heavily dependent on the availability and uptime of the payment gateway for processing financial transactions and Google Maps or similar geospatial APIs for visualizing service areas. Furthermore, the accuracy of the system's inventory and subscription data depends on the consistent and timely updates provided by the administrative and field staff. It is also assumed that the underlying server environment, whether on-premises or cloud-based, provides sufficient resources to handle the concurrent requests from administrators, staff, and the customer base.

3.3 Specific Requirements

The Specific Requirements define the technical and functional backbone of the system. External Interfaces ensure seamless interaction between the React frontend, Django backend, and third-party services like Google Maps. The Functional Requirements automate the entire ISP lifecycle, from capturing leads to automated invoicing and smart support ticketing with staff routing. To maintain a professional user experience, Performance Requirements mandate sub-500ms API response times and high system availability, especially during peak billing periods. The Logical Database Requirements enforce a normalized relational schema that secures sensitive customer and financial data while supporting geospatial zone mapping. The design is further guided by Constraints such as the decoupled architecture and JWT security, while System Attributes like maintainability and reliability ensure long-term stability.

3.3.1 External Interfaces

The NextGenISP system interacts with several external entities through well-defined software and hardware interfaces. The user interfaces provide distinct web-based dashboards for administrators, staff, and customers, all developed using responsive React components to ensure compatibility across various devices. The software interfaces include critical integrations with the Razorpay API for processing financial transactions and geospatial mapping APIs for visualizing network service areas. Communication between the React frontend and the Django backend is managed through a RESTful API using the JSON format for data exchange. On the hardware side, the system's management capabilities interface indirectly with network infrastructure devices, such as routers and ONTs, by tracking their MAC addresses and serial numbers within the central database.

3.3.2 Functional Requirements

The functional requirements of NextGenISP are centered on providing a seamless, automated workflow for managing the ISP's core business operations. This includes the ability for the system to capture and track prospective leads from the public website and transition them through the various stages of the installation and activation pipeline. The system must also support the automated generation of monthly recurring invoices and provide a secure mechanism for customers to view and pay their bills online. Furthermore, the CRM module must allow for the creation and management of technical support tickets, with features for categorizing issues and assigning tasks to the appropriate field or technical staff. The inventory management functions correctly track the allocation and return of hardware assets, while the GIS mapping functionality provides the ability for administrators to visually manage service zones and monitor network coverage.

3.3.3 Performance Requirements

To ensure a high-quality user experience, the system must adhere to several key performance standards regarding responsiveness and reliability. The web-based dashboards are expected to load within an acceptable timeframe, with API response times for standard data retrieval operations targeted at under 500 milliseconds. The system's backend should be capable of handling multiple concurrent requests from administrators, staff, and customers without a significant degradation in performance. For critical financial operations, such as invoice generation and payment processing, the system must ensure data consistency and high availability, particularly during the peak billing periods at the start of each month. Scalability is also a vital requirement, as the platform must maintain its performance levels as the number of active subscribers and network nodes grows over time.

3.3.4 Logical Database Requirements

The logical database architecture of NextGenISP is designed to ensure maximum data integrity and consistency through a highly structured relational schema.

- **Normalization Level:** The database is structured up to the Third Normal Form (3NF) to minimize data redundancy and eliminate potential anomalies during insertion, updating, and deletion.
- **Database System:** The application uses SQLite for development and is fully compatible with PostgreSQL or MySQL for high-load production environments.
- **Geospatial Data:** The database supports the storage of complex JSON-formatted coordinate polygons to represent service zones (Areas), allowing for efficient zone-based queries.
- **Data Security:** Strict Role-Based Access Control (RBAC) is enforced at the database layer to ensure that users can only query or modify data permitted by their specific role (Admin, Staff, or Customer).

3.3.5 Design Constraints

The design of the NextGenISP is influenced by several technical and operational constraints that guide its development. The system is built using a decoupled architecture mandated by the selection of the Django REST Framework and React, which necessitates a strict separation between the API layer and the user interface. Security is a primary design constraint, requiring the implementation of JWT-based authentication and HTTPS encryption for all data transmissions. Furthermore, the choice of a relational database system like SQLite or PostgreSQL imposes specific data modeling constraints to maintain normalization and performance, particularly when handling geospatial data and large-scale billing records.

3.3.6 Software System Attributes

NextGenISP is characterized by several critical software system attributes, including reliability, maintainability, and security. Reliability is ensured through robust error handling and the use of stable, well-supported frameworks for both the backend and frontend. The system's maintainability is achieved through a modular code structure and clear separation of concerns, which allows for independent updates to specific business modules without affecting the entire platform. Security is prioritized through the use of role-based authorization, ensuring that sensitive financial and personal data is only accessible to authorized personnel. Additionally, the system's portability is maintained by using cross-platform technologies like Python and JavaScript, allowing the ERP to be deployed on various server environments with minimal configuration changes.

3.3.7 Organizing the Specific Requirements

In this system the overall functionality is organized by Data flow diagrams and of E-R diagrams. Based on these diagrams, data relationships and dependencies are found and a functional hierarchy is made for organizing the specific requirements.

4. SOFTWARE AND HARDWARE REQUIREMENT

4.1 Software Requirements

This section outlines the essential technical specifications required for the development, deployment, and operation of the NextGenISP system. The minimal software's used for the storage are :

- Operating Systems: Windows 10/11 (Development/Client),Linux Ubuntu 20.04+ (Deployment/Server).
- Programming Languages: Python 3.10+ (Backend), JavaScript / JSX (Frontend)
- Backend Framework: Django 5.0 and Django REST Framework (DRF)
- Frontend Library: React 18+ (bundled with Vite)
- Database Systems: Development: SQLite (Relational)Production: PostgreSQL or MySQL

4.2 Hardware Requirements

The hardware requirements for the NextGenISP platform outline the necessary specifications and configurations that users' computer systems must meet to ensure optimal performance of the software. Below is a specification of how these hardware requirements are defined:

- Processor: Dual-core processor, 2.0 GHz or equivalent
- Memory (RAM): 8 GB or higher
- Storage: Minimum 20 GB of available disk space
- Display: 1280x800 resolution or higher
- Input Devices: Keyboard and mouse (or other pointing devices)

5. SYSTEM DESIGN

5.1 Introduction

The system design phase for the NextGenISP focuses on transforming the established requirements into a detailed technical blueprint. This section outlines the architectural framework, data models, and interface designs that govern how the system functions the system design ensures a robust, scalable, and secure environment that can effectively handle the complex operational needs of a modern internet service provider.

5.2 Input Design

Join NextGen ISP

Fast & Reliable Internet Connection

Check Availability

Select your area to see if we are available.

Select Location on Map (Recommended)

Show Map

Select Area Manually

-- Choose your location --

Select Area to Proceed →

Already have an account? Login

Join NextGen ISP

Fast & Reliable Internet Connection

FIRST NAME

John

LAST NAME

Doe

USERNAME

john Doe123

EMAIL

john@example.com

PHONE

+91 9876543210

ADDRESS

House No, Street, Landmark...

ID PROOF (GOVERNMENT ID)

Click to upload ID Proof

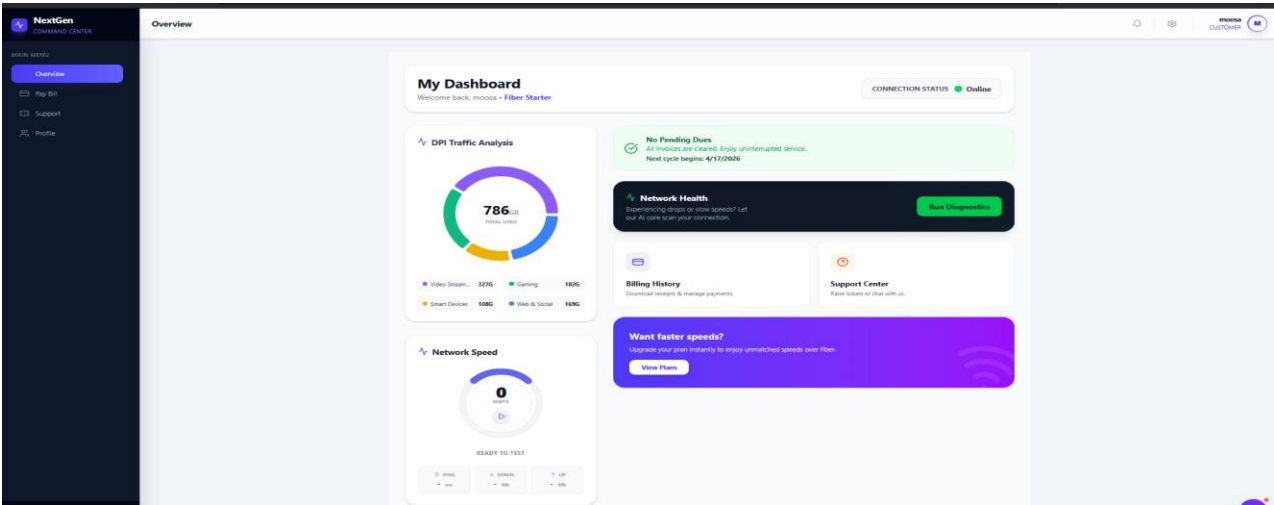
PASSWORD

CONFIRM

Register for Connection

Back to Location Check

5.3 Output Design



5.4 Module Description

The NextGenISP system is organized into six core integrated modules, each designed to handle a critical aspect of an ISP's business operations. These modules work in synchronization to provide a unified work experience.

- **User and Identity Management Module**

This module is the security foundation of the system, managing Role-Based Access Control (RBAC) across four primary user groups: Admins, Technical Staff, Field Staff, and Customers. It handles secure user authentication via JWT. It also manages the global state for user status, ranging from prospective leads to active subscribers or suspended accounts.

- **Operations and Onboarding Module**

The Operations module manages the end-to-end customer lifecycle, beginning from initial public enquiries through the automated lead verification process. It facilitates the transition of verified leads into the installation pipeline, where tasks are assigned to field and technical staff for physical setup and logical configuration. Once the installation is confirmed, this module triggers the service activation, moving the customer from a pending status to an active service subscription.

- **Billing and Financial Module**

This module automates the entire financial cycle of the ISP, reducing manual administrative labor and ensuring consistent revenue. It involves a recurring billing engine that generates automated invoices based on the customer's selected internet plan and billing cycle. It processes online transactions in real-time and automatically updates the customer's subscription status upon successful payment, reducing the risk of revenue leakage through manual errors.

- **Service and Support (CRM) Module**

The CRM module provides a comprehensive technical support ecosystem for both customers and staff. It enables customers to raise support tickets for both physical (e.g., fiber breakage) and logical (e.g., speed issues) problems. The module facilitates a smart ticketing workflow where issues are categorized and assigned to the appropriate staff members for resolution, while providing a real-time communication channel through integrated chat history between the support team and the end-user.

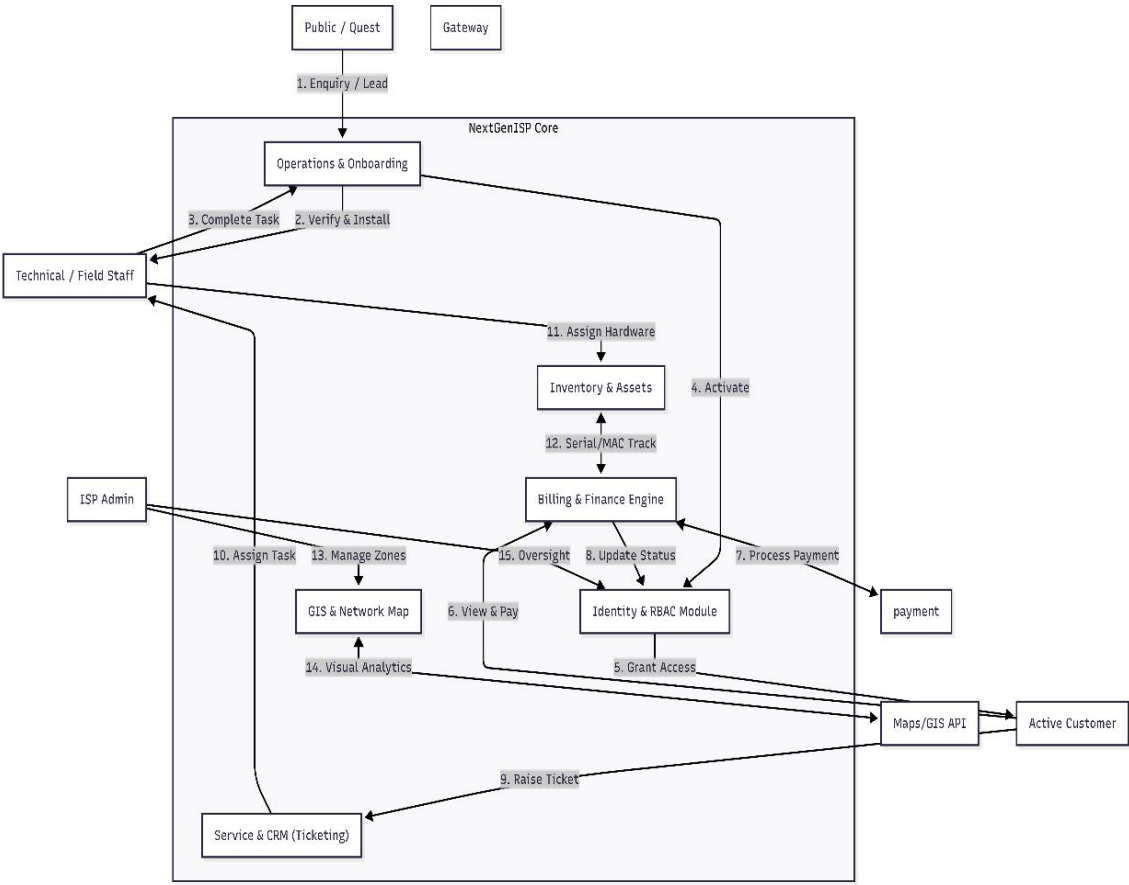
- **Geographic Information System (GIS) Module**

The GIS module provides a geospatial layer to the ISP's management functions, allowing administrators to visualize service coverage across different geographic zones (Areas). Using coordinate-based polygons, it maps out active customer clusters and service availability, enabling the ISP to make data-driven decisions regarding network expansion. It also provides a visual interface for managing localized maintenance alerts and network outage monitoring across specific service areas.

- **Inventory and Asset Management Module**

This module is responsible for tracking the lifecycle of critical hardware assets, such as routers and ONTs (Optical Network Terminals). It maintains a master inventory of available stock and tracks the precise allocation of hardware to specific customers or staff members using unique serial numbers and MAC addresses. By ensuring that every physical asset is precisely tracked and accounted for, the module helps minimize equipment loss and improves the overall capital efficiency of the ISP.

5.5 Functional Diagram



5.6 Database Design

api_area

Field	Type	Constraint	Description
ID	BigInt	PK	Unique System Identifier
name	Varchar(100)	Not null	Name of area
code	Varchar(20)	Unique	Unique code
city	Varchar(100)	Default	City location
coordinates	Text	Null	Json list
Is_under_maintenance	Boolean	Default	Global maintenance toggle

Api_plan

Field	Type	Constraint	Description
Id	Bigint	PK	Unique system identifier
Name	Varchar(100)	Not null	Plan name
Speed_mbps	Integer	Not null	Connection speed
Price	Decimal	Not null	Monthly cost
Plan_type	Varchar(10)	Choice	Fiber/wireless

Api_hardware

Field	Type	Constraint	Description
Id	Bigint	PK	Unique Identifier
name	Varchar(100)	Not null	Device name
price	Decimal	Not null	Retail price

Api_user

Field	Type	Constraint	Description
Id	Bigint	PK	Unique identifier
Username	Varchar(150)	Unique	Login name
Role	Varchar(20)	Choice	Admin,staff,custom er
Area_id	BigInt	Fk(api_area)	Assigned service zone
Status	Varchar(20)	Choice	Lead,active,suspend ed

Api_subscription

Field	Type	Constraint	Description
Id	Bigint	PK	Unique Identifier
User_id	Bigint	FK(api_user)	Linked customer
Plan_id	Bigint	FK(api_user)	Selected internet plan
status	Varchar(10)	choice	Active,expired,suspened

Api_invoice

Field	Type	Constraint	Description
Id	Bigint	PK	Unique Identifier
Subscription_id	Bigint	FK(api_sub)	Source subscription
Amount	Decimal	Not null	Total billed amount
Status	Varchar(10)	Choice	Pending,Paid,overdue
Due_date	Date	Not null	Payment deadline

Api_payment

Field	Type	Constraint	Description
Id	BigInt	PK	Unique identifier
Invoice_id	Bigint	FK(api_invoice)	Linked invoice
Transaction_id	Varchar(100)	Unique	Gateway transaction id
method	Varchar(50)	Default	Online or cash

Api_ticket

Field	Type	Constraint	Description
Id	BigInt	PK	Unique Identifier
Customer_id	Bigint	Fk(api_user)	User who raised ticket
Assigned_to_id	Bigint	Fk(api_user)	Staff handling issue
status	Varchar(20)	Choice	Open,resolved,closed

Api_installationtask

Field	type	Constraint	Description
Id	Bigint	PK	Unique identifier
Customer_id	Bigint	FK(api_user)	Customer being installed
Assigned_staff_id	BigInt	FK(api_user)	Assigned field tech
status	Varchar(20)	Choice	Pending,inprogress,closed

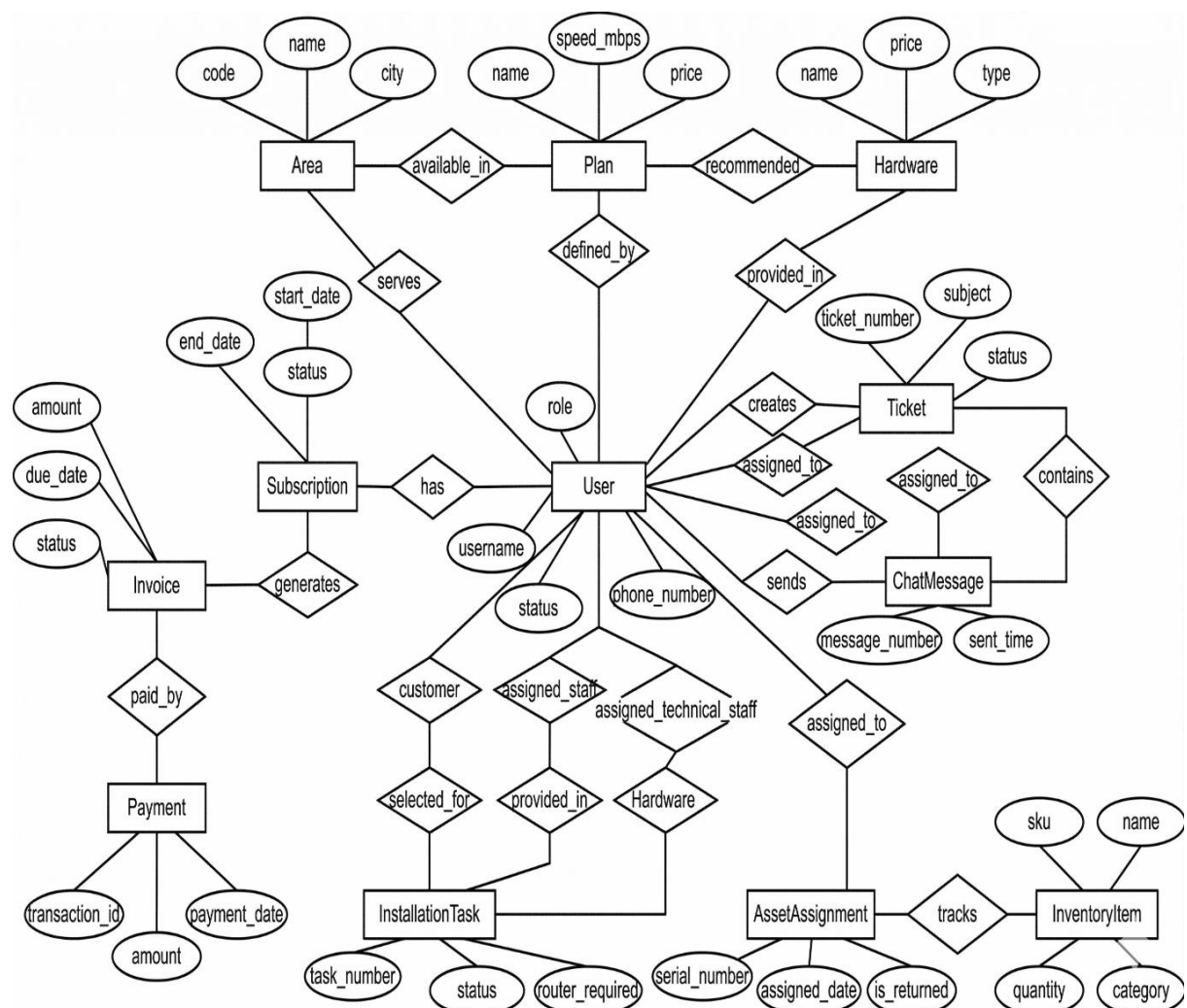
Api_inventoryitem

Field	Type	Constraint	Description
Id	Bigint	PK	Unqiue identifier
Sku	Varchar(50)	Unique	Stock keeping unit
quantity	Integer	Deafult	Current warehouse stack

Api_assetassignment

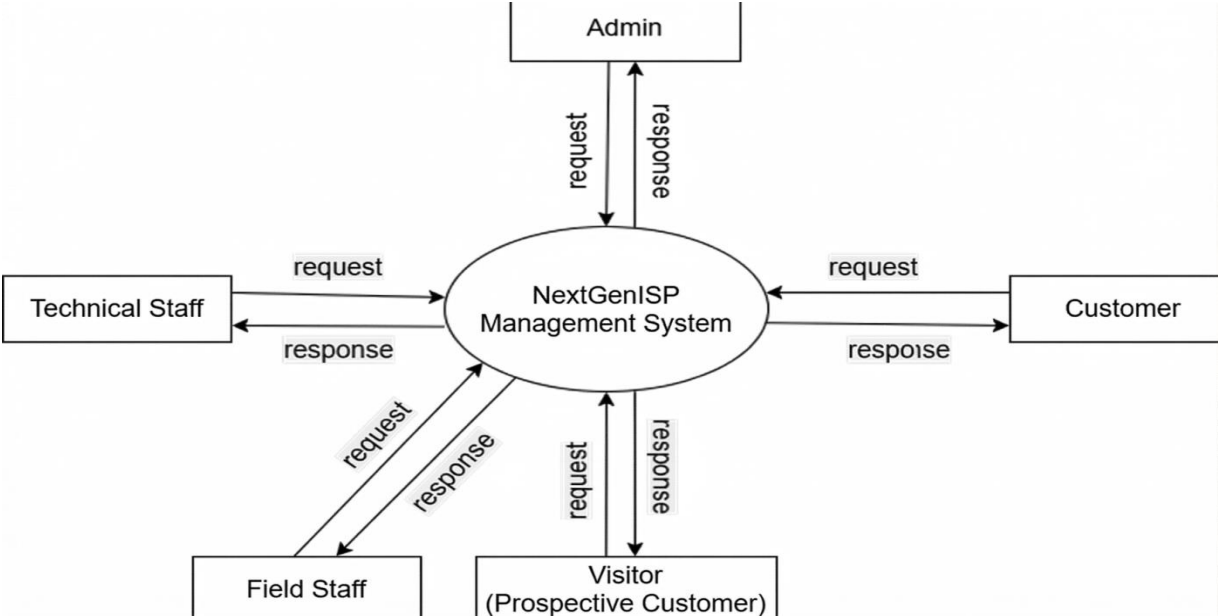
Field	Type	Constraint	Description
Id	Bigint	PK	Unique identifier
Item_id	Bigint	FK(api_inv)	Type of hardware issued
User_id	Bigint	FK(api_user)	Holder
Serial_number	Varchar(100)	Not null	Specifi device id

5.7 ER Diagram

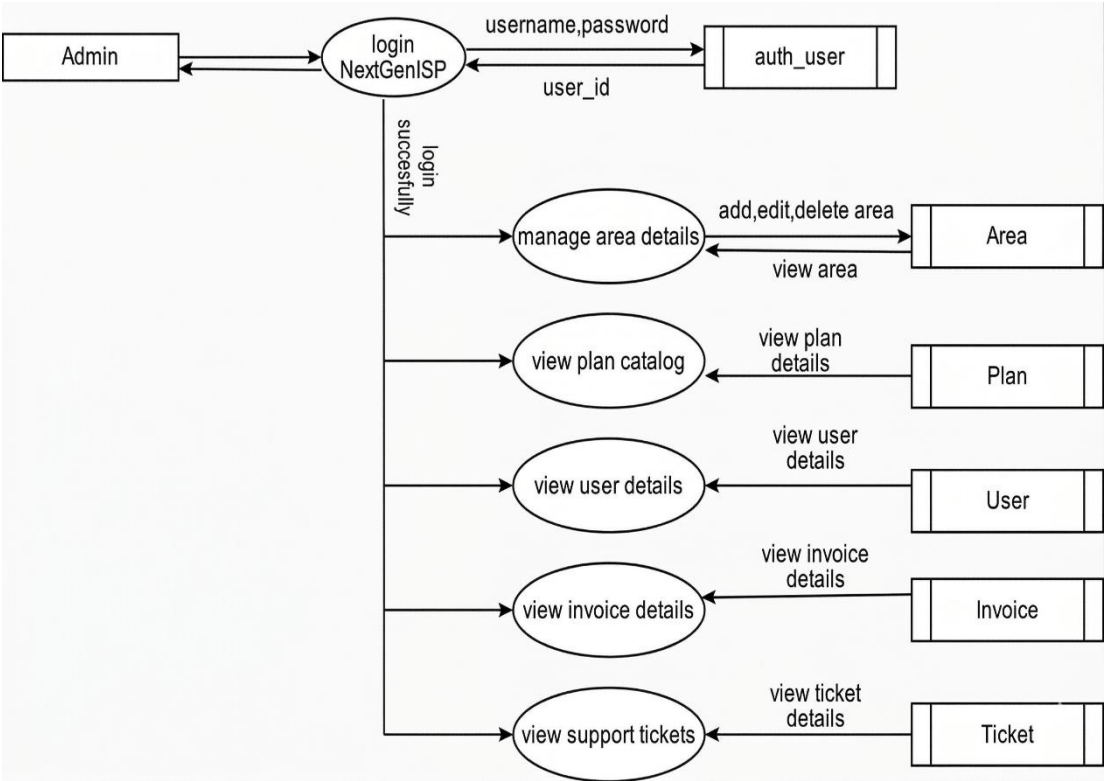


5.8 Data Flow Diagram

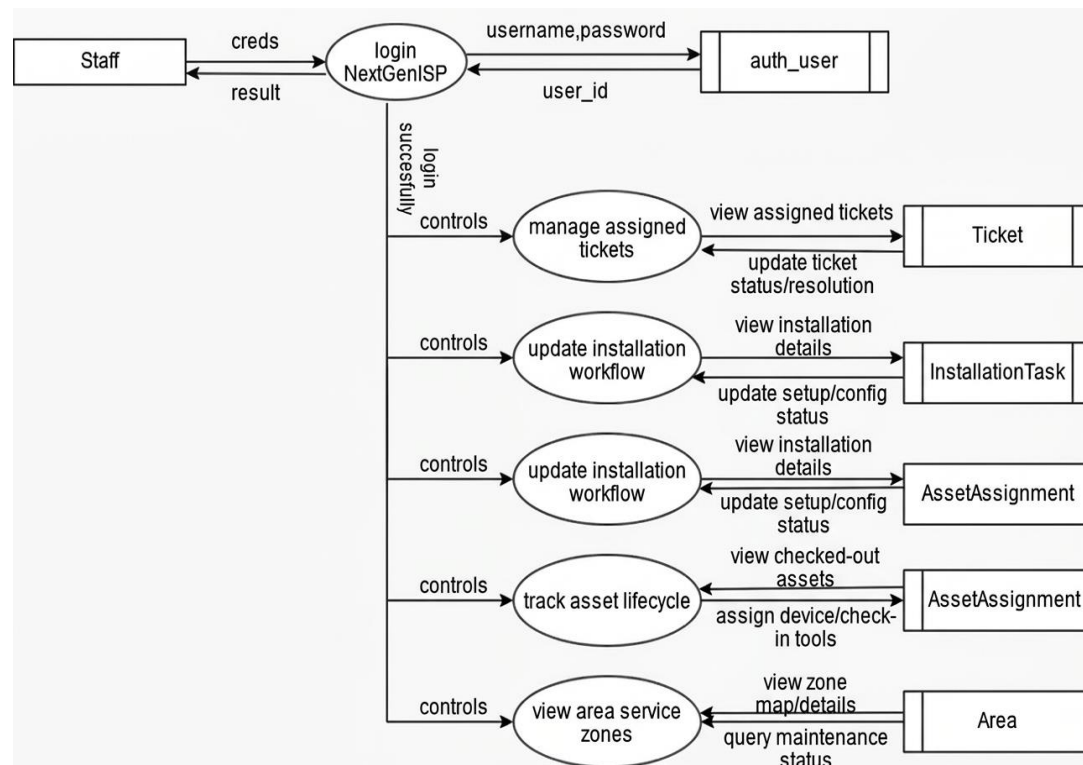
Level 0



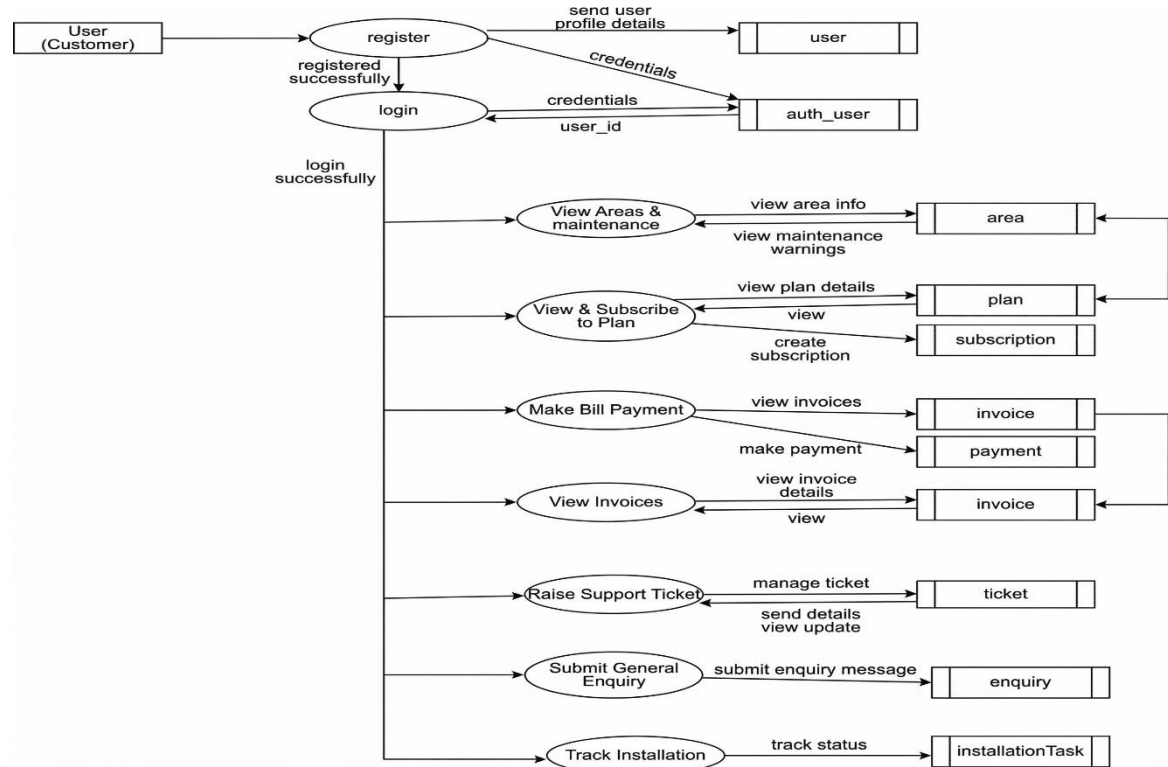
Level 1 Admin



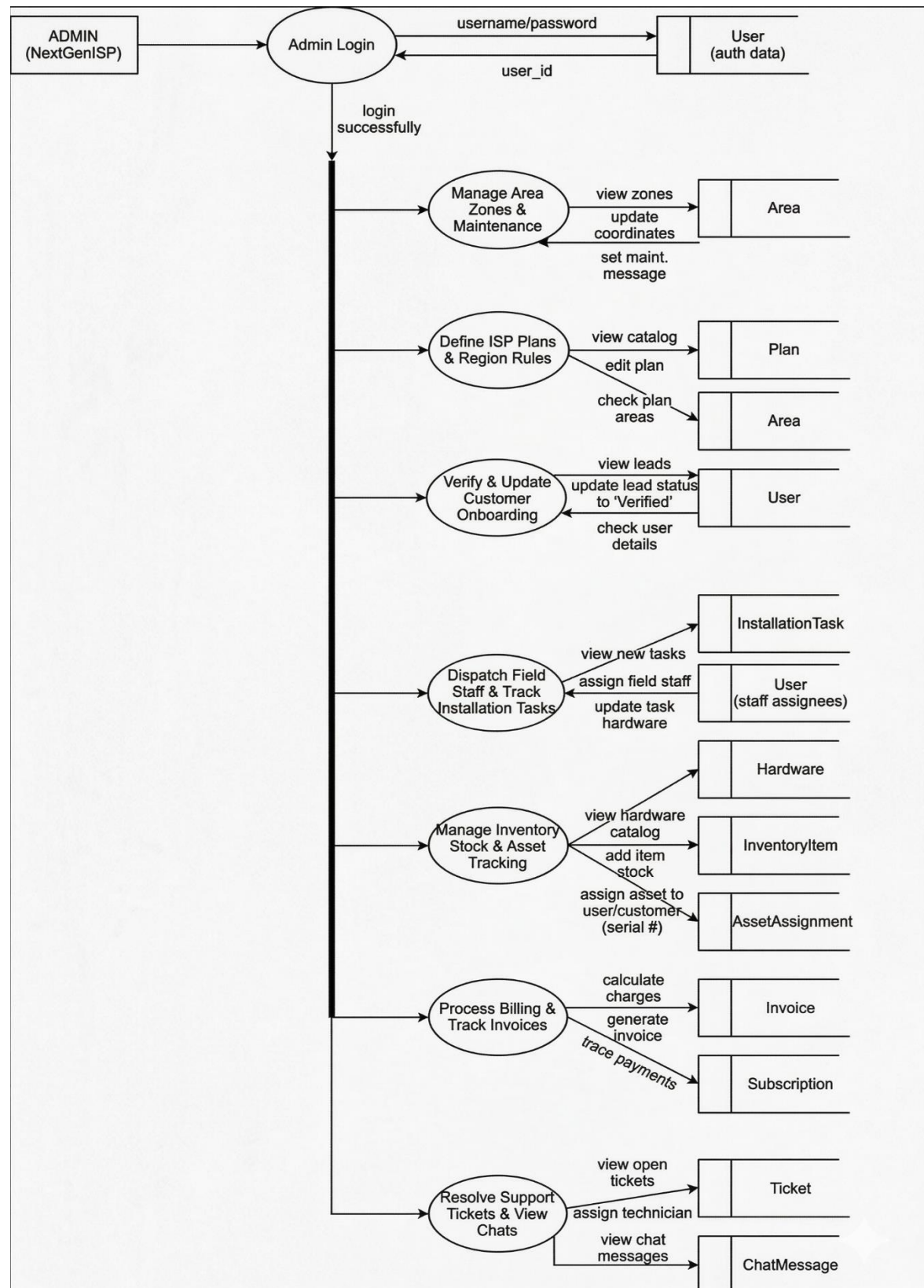
Level 1 Staff



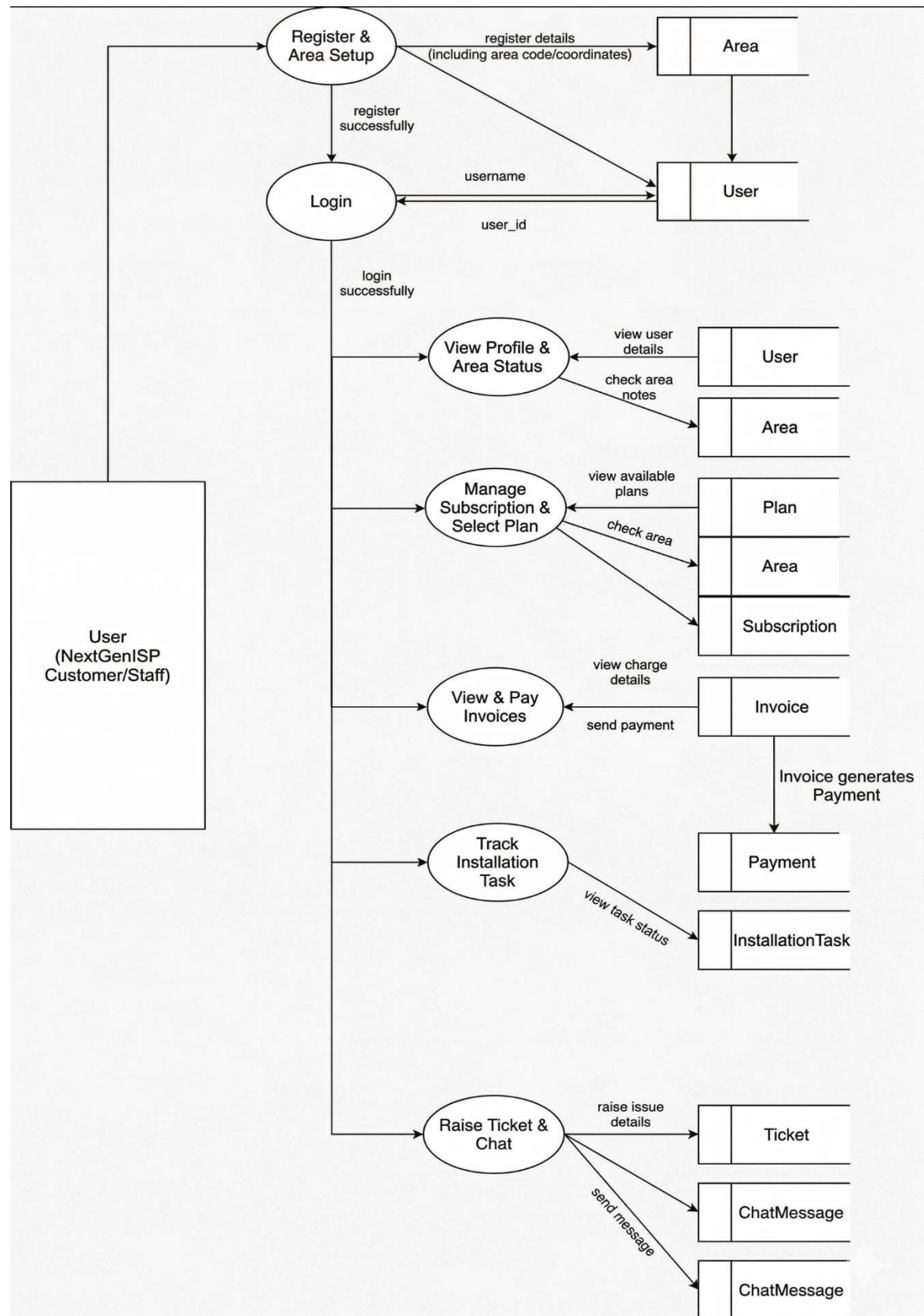
Level 1 User



Level 2 Admin



Level 2 User



6. SYSTEM DEVELOPMENT

6.1 Process Description

The development of the NextGenISP system follows an iterative and incremental process, grounded in Agile methodology to ensure flexibility and alignment with evolving business requirements. The lifecycle begins with comprehensive requirement analysis and data modeling, which informs the creation of the Django-based REST API. Development is segmented into logical modules, such as user identity, billing, and geographic information systems, allowing for focused implementation and unit testing of core business logic. As the backend services are established, the React-based frontend is developed in parallel, utilizing modern component-driven patterns to ensure a high-performance and responsive user experience. The process emphasizes continuous integration, where incremental updates are frequently tested against the actual database schema and functional workflows to ensure seamless communication between the client and server. This collaborative development cycle concludes with thorough system verification and cross-browser testing to ensure the platform is robust, secure, and ready for real-world ISP operations.

6.2 Psuedo Code

The algorithmic logic for the NextGenISP ensures that complex operational tasks, such as automated service activation and recursive billing, are handled with high precision and minimal human intervention. Below are the pseudo-code representations for two core system workflows.

- **Customer Onboarding and Service Activation**

This algorithm governs the transition of a verified lead into an active subscriber, coordinating between physical installation and logical network configuration.

```
FUNCTION ActivateCustomerService(LeadID, AssignedStaffID)
```

```
    // 1. Initial Validation
```

```
    SET Lead = GET CustomerData FROM api_user WHERE id = LeadID
```

```
    IF Lead.status != 'READY_TO_INSTALL' THEN
```



```

    RETURN "Error: Lead not verified for installation"

END IF

// 2. Physical Layer: Installation Task Creation

CREATE Task IN api_installationtask(customer=Lead, staff=AssignedStaffID,
status='PENDING')

// Await field signal for hardware mounting and fiber termination

WHILE Task.status != 'PHYSICAL_COMPLETED'

    AWAIT Signal FROM MobileApp(LeadID)

    IF Signal.Success THEN

        UPDATE Task SET status = 'PHYSICAL_COMPLETED'

        RECORD HardwareDetails(Lead.hardware_id, router_mac_address)

    END IF

END WHILE

// 3. Logical Layer: Automated Provisioning

ASSIGN Dynamic_IP(router_mac_address)

IF RouterReachabilityTest(router_mac_address) IS SUCCESSFUL THEN

    // 4. Final Activation

    UPDATE api_user SET status = 'ACTIVE' WHERE id = LeadID

    CREATE api_subscription(user=Lead, plan=Lead.plan, status='ACTIVE',
start=TODAY)

    UPDATE Task SET status = 'CLOSED'

    SEND_NOTIFICATION(Lead, "Welcome! Your high-speed internet is now
active.")

ELSE

    LOG_ERROR("Logical Provisioning Failed - Check OLT/ONT Handshake")

    SEND_ALERT(TechSupport, "Urgent: Configuration error for User " +
Lead.username)

```

END IF

END FUNCTION

- **Automated Billing Engine and Subscription Suspension**

This recurring background process handles the financial lifecycle of active subscriptions, managing automated invoice generation and service suspension for non-payment.

FUNCTION RunDailyBillingCycle()

// Identify all subscriptions requiring action

SET ActiveSubs = GET ALL FROM api_subscription WHERE status = 'ACTIVE'

FOR EACH Sub IN ActiveSubs

// Step 1: Check for Expiry

IF CURRENT_DATE >= Sub.end_date THEN

// Step 2: Automated Invoice Generation

CREATE api_invoice(

subscription = Sub,

amount = Sub.plan.price,

due_date = CURRENT_DATE + 5 DAYS,

status = 'PENDING')

// Step 3: Evaluate Suspension Grace Period

// Suspend service if payment is 3 days overdue post-expiry

IF CURRENT_DATE > (Sub.end_date + 3 DAYS) THEN

UPDATE api_subscription SET status = 'SUSPENDED' WHERE id = Sub.id

UPDATE api_user SET status = 'SUSPENDED' WHERE id = Sub.user_id

// Trigger Hardware Access Block via Network API

```
        SEND_API_COMMAND(NetworkController, "BLOCK_MAC",  
Sub.user.router_mac)
```

```
        SEND_SMS_ALERT(Sub.user, "Service suspended due to pending dues.  
Repay now to restore.")
```

```
    ELSE
```

```
        SEND_EMAIL_REMINDER(Sub.user, "Your subscription has ended. Pay  
your pending invoice within 3 days.")
```

```
    END IF
```

```
END IF
```

```
END FOR
```

```
END FUNCTION
```

7. VALIDATION CHECKS

Validation checks in the NextGenISP ERP system are integrated at multiple layers to ensure data integrity, system security, and operational reliability:

- **Client-Side Validation:** The React frontend implements real-time form validation to provide immediate feedback on user inputs, such as telephone numbers, email addresses, and MAC address formats, preventing erroneous data from reaching the server.
- **Server-Side Schema Validation:** On the backend, the Django REST Framework utilizes robust serializers to enforce strict schema validation, ensuring that all incoming API requests adhere to the expected data types and mandatory field requirements.
- **Business Logic Verification:** The system implements complex business logic validation, such as verifying the availability of specific internet plans within a geographic zone before allowing a subscription to proceed, and cross-referencing hardware serial numbers against existing inventory.
- **Security & Authorization Checks:** Security validation is prioritized through JWT-based token verification and role-based access checks (RBAC) on every request, ensuring that sensitive administrative and financial operations are only performed by authorized personnel.
- **Data Integrity Constraints:** Multi-layered protocols collectively maintain a high standard of data consistency at the database level, protecting the platform against both accidental errors and malicious interactions.

8. SYSTEM IMPLEMENTATION

8.1 Testing

Testing is one of the most important stages in the software development life cycle. It ensures that the developed system works as expected and that all modules function correctly before deployment. The NextGenISP System was thoroughly tested to verify its functionality, performance, and reliability. The purpose of testing was to detect and eliminate errors, validate data accuracy, and confirm that the system met all specified requirements. The testing process began after the successful completion of system development and integration. Both individual module testing and overall system testing were carried out systematically. The key testing phases included:

1. **Unit testing:** Unit testing was conducted on individual components such as the login module, customer onboarding, billing engine, and geographic area management sections.
2. **Integration testing:** After unit testing, integration testing was performed to ensure that the different modules of the system, such as the CRM ticketing and hardware inventory, worked together correctly as a unified ERP.
3. **Functional Testing:** Functional testing focused on checking whether each feature of the system worked according to its intended purpose. All major functions such as secure login, subscription creation, invoice generation, and real-time network map visualization were tested with sample customer data.
4. **Validation and Input Testing:** This testing ensured that user inputs were properly validated and restricted. Incorrect or incomplete data entries, such as invalid phone numbers, incorrect MAC addresses, or improperly formatted geospatial coordinates, were used to verify that the system handles errors gracefully.
5. **Security Testing:** Since the system deals with confidential financial and personal customer information, security testing was carried out to verify that only authenticated users with specific roles (Admin/Staff/Customer) could access their respective modules.

6. **Performance Testing:** Performance testing was done to evaluate the response time and stability of the system under different loads. Multiple simultaneous logins and real-time data operations, such as simultaneous invoice generation for hundreds of users, were performed to ensure the system could handle them without slowdowns or crashes.
7. **User Acceptance Testing:** After completing all technical testing, the system was presented to end users (administrators and officers) for User Acceptance Testing. They used the system in real scenarios to verify that it met their daily operational needs.

8.2 System Implementation

The implementation of the NextGenISP involves the systematic deployment and configuration of the platform within a production environment. This phase includes the setup of the primary database system (e.g., PostgreSQL), the configuration of the web server and application hosting environment, and the final migration of the validated codebase. Implementation is conducted in stages, beginning with a pilot deployment in a controlled environment to verify stability and performance under simulated production loads. Once verified, the full system is launched, accompanied by the initialization of essential master data, such as service area coordinates and internet plan specifications. The implementation process also prioritizes user training and documentation to ensure that all roles can effectively utilize the system's features from the first day of operation.

8.3 Security

Security is a foundational pillar of the NextGenISP implemented through a multi-layered strategy designed to protect sensitive financial and personal data. The system utilizes industry-standard JSON Web Tokens (JWT) for secure, stateless authentication, ensuring that all API interactions are validated against a verified user session. Role-Based Access Control (RBAC) is strictly enforced at every architectural layer, restricting data access and administrative functions to authorized personnel based on their predefined roles. Furthermore, all data transmissions are encrypted using HTTPS to prevent interception, and the system implements robust input sanitization and secure query practices to protect against common web vulnerabilities such as SQL injection and cross-site scripting (XSS). Regular security audits and proactive monitoring of system logs are also integrated into the operational lifecycle to ensure ongoing protection against emerging threats.

9. FUTURE SCOPE OF THE PROJECT

The NextGenISP system is designed with a modular architecture that allows for significant future expansion and the integration of emerging technologies. The roadmap for future enhancements focuses on increasing automation, improving network reliability, and enhancing the overall customer experience through the following initiatives:

- **AI-Driven Predictive Maintenance:** Future versions will integrate machine learning algorithms to analyze network traffic patterns and hardware telemetry data. This will allow the system to predict potential equipment failures or fiber cuts before they affect service, automatically generating preemptive maintenance tickets for field staff.
- **Advanced Network Monitoring (SNMP Integration):** Direct integration with OLT (Optical Line Terminal) and ONT (Optical Network Terminal) hardware via SNMP protocols will provide real-time, port-level visibility. This will enable automated logical troubleshooting and instant detection of physical layer issues without manual intervention.
- **Dedicated Field Staff Mobile Application:** To further empower field technicians, a native mobile application with offline capabilities and real-time GPS tracking will be developed. This will facilitate more efficient task management, digital signature capture for installations, and optimized route planning based on geographic service zones.
- **IoT and Smart Home Service Models:** The system will be expanded to support emerging Internet of Things (IoT) subscriptions, allowing the ISP to manage and bill for value-added services such as smart home security monitoring and managed Wi-Fi ecosystems.
- **Enhanced Customer Self-Service (AI Chatbots):** The integration of advanced natural language processing (NLP) will lead to the development of sophisticated AI chatbots within the customer portal. These bots will handle routine enquiries, process basic plan.

9. CONCLUSION

The development of the NextGenISP system represents a significant step forward in automating and streamlining the complex operations of an internet service provider. By integrating core functional modules—ranging from identity management and billing to geographic information systems and inventory tracking—into a unified, role-based platform, the system successfully addresses the challenges of manual administration and data fragmentation.

The transition to a modern, decoupled architecture using Django and React has not only improved the overall performance and responsiveness of the platform but has also provided a scalable foundation for future technical enhancements. Through rigorous validation, testing, and a focus on security, NextGenISP ensures that essential business processes are handled with accuracy and reliability, ultimately enabling the ISP to deliver a more efficient and professional service to its customers. The project stands as a robust solution that is well-prepared to adapt to the evolving demands of the telecommunications industry.

11. APPENDIX

11.1 Sample Inputs

Join NextGen ISP

Fast & Reliable Internet Connection

FIRST NAME

John

LAST NAME

Doe

USERNAME

john Doe123

EMAIL

john@example.com

PHONE

+91 9876543210

ADDRESS

House No, Street, Landmark...

ID PROOF (GOVERNMENT ID)

Click to upload ID Proof

PASSWORD

CONFIRM

Register for Connection

Back to Location Check

Join NextGen ISP

Fast & Reliable Internet Connection

Check Availability

Select your area to see if we are available.

Select Location on Map (Recommended)

Show Map

Select Area Manually

-- Choose your location --

Select Area to Proceed

Already have an account? Login

11.2 Sample Outputs

NextGen

War Room

Network Operations Center

Online

Successful

100%

ZONE METRICS

Total Nodes

Critical Alerts

Network Health 99.9%

NextGen

Models / Plan

Manage Offerings

Internet Plans

Hardware

Plan Name

Speed (Mbps)

Price (₹)

Type

Available Area

Create Plan

Fiber Starter

₹499.00/month

50 Mbps Speed

Available in 2 zones

Fiber Gamer

₹999.00/month

150 Mbps Speed

Available in 1 zones

Gamer Pro 500

₹1499.00/month

500 Mbps Speed

Available in 2 zones

Business Elite 1000

₹2999.00/month

1000 Mbps Speed

Available in 1 zones

Family Starter

₹499.00/month

50 Mbps Speed

Available in 2 zones

Fiber Intermediate

₹400.00/month

40 Mbps Speed

Available in 1 zones

NextGen ISP

Plans & PricingWhy UsCustomerStaffAdmin

THE FUTURE OF CONNECTIVITY

Internet that moves as Fast as You

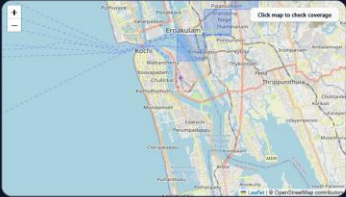
Experience ultra-low latency, 99.9% uptime, and dedicated support. Join thousands of happy subscribers today.

[Check Availability](#)[Use My Location](#)[View Plans](#)

We are expanding fast!

Check if NextGen Fiber is available in your area. Click on the map to verify coverage instantly.

Click map to check coverage



NextGen ISP

Plans & PricingWhy UsCustomerStaffAdmin

HARDWARE SOLUTIONS

Certified Devices for Max Speed

Choose the right gear for your plan. One-time purchase, yours to keep.



Basic ONU Wi-Fi Modem
₹ 1200.00
Premium Performance & Ultimate Reliability
[View Specifications](#)



Dual-Band Gigabit Router
₹ 1499.00
The Perfect Balance of Speed and Value
[View Specifications](#)



Wi-Fi Range Extender
₹ 1500.00
Premium Performance & Ultimate Reliability
[View Specifications](#)



TP-LINK TL-WR820N
₹ 2500.00
Your Fast, Everyday Wi-Fi Essential
[View Specifications](#)



WiFi 6 Mesh Router
₹ 3499.00
Premium Performance & Ultimate Reliability
[View Specifications](#)



Gaming Router Pro
₹ 3999.00
Premium Performance & Ultimate Reliability
[View Specifications](#)

11.3 Reports

NextGen ISP - Comprehensive System Report

Generated on: 2026-03-25 07:26:13

1. Executive Summary

Metric	Count / Value
Total Customers	14
Total Staff Members	7
Pending Installations	3
Open Support Tickets	4
Total Revenue Collected	INR 11494.00

2. Internal Staff List

ID	Name	Role	Joined
6	admin	ADMIN	2026-01-31
19	Tech_manu	TECHNICAL_STAFF	2026-03-04
20	Field_gopu	FIELD_STAFF	2026-03-05
25	test_staff_2	TECH_STAFF	2026-03-18
28	sim_staff2	TECH_STAFF	2026-03-18
30	sim_staff99	TECH_STAFF	2026-03-18
32	sim_staff100	TECH_STAFF	2026-03-18

3. Recent Customer Registrations

ID	Username	Email	Area	Status
31	sim100	sim100@sim.com	N/A	ACTIVE
29	sim99	sim99@sim.com	N/A	ACTIVE
27	sim22	sim22@sim.com	N/A	ACTIVE
26	brand_new_sim	brandnew@sim.com	N/A	VERIFIED
22	anzalna	anzalnayusaf@gmail.com	Tech Park Phase 2	ACTIVE
21	moosa	vavisahad76@gmail.com	Edapally Zone	ACTIVE
17	carlos	carlos@gmail.com	Uptown Business Distri	ctACTIVE
16	paul	paul@example.com	N/A	VERIFIED

15	ravi	ravi@example.com	N/A	VERIFIED
14	melisa	melisa@gmail.com	Kochi Central	INSTALLATION_PENDING

4. Recent Plan Upgrades

Date	User	New Plan	Amount Paid
2026-03-25	moosa	Business Elite 1000	INR 2500.00
2026-03-18	anzalna	Fiber Gamer	INR 500.00
2026-03-18	anzalna	Corporate Fiber	INR 4500.00

5. Open Support Tickets

ID	Subject	Type	Status	User
7	Issue regarding Speed	LOGICAL	RESOLVED	cust_bob
5	Issue regarding Connectio	LOGICAL	OPEN	cust_sarah
4	Issue regarding Speed	INSTALLATION	OPEN	cust_bob
2	Issue regarding Speed	LOGICAL	IN_PROGRESS	cust_bob

6. Recent Bill Payments

Date	Transaction ID	User	Amount	Method
2026-03-25	SIM_TXN_20_UPI	moosa	INR 2500.00	UPI
2026-03-18	SIM_TXN_19_UPI	anzalna	INR 4500.00	UPI
2026-03-18	SIM_TXN_17_UPI	anzalna	INR 500.00	UPI
2026-01-28	TXN-88646	cust_sarah	INR 999.00	ONLINE

7. Recent Customer Enquiries

Date	Name	Subject	Status
2026-03-18	sim100	Plan Selection & Installa	OPEN
2026-03-18	sim99	Plan Selection & Installa	OPEN
2026-03-18	sim22	Plan Selection & Installa	OPEN
2026-03-12	anzalna	Plan Selection & Installa	OPEN
2026-03-05	moosa	Plan Selection & Installa	OPEN
2026-03-04	moosa	Plan Selection & Installa	OPEN
2026-03-04	moosa	Plan Selection & Installa	OPEN
2026-03-04	moosa	Plan Selection & Installa	OPEN
2026-03-04	moosa	Plan Selection & Installa	OPEN

2026-03-04	moosa	Plan Selection & Installa	OPEN
------------	-------	---------------------------	------

--- End of Report ---

NextGen ISP

Customer Bill

INVOICE #19

Date: 2026-03-18
Status: PAID

Bill To:

anzalna
anzalnayusaf@gmail.com

9072189290

Description	Plan Details	Amount
Broadband Subscription	Fiber Starter	INR 4500.00
	CGST + SGST (18%)	INR 810.00
	Total Payable	INR 5310.00

Thank you for your business!

This is a computer-generated invoice.

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